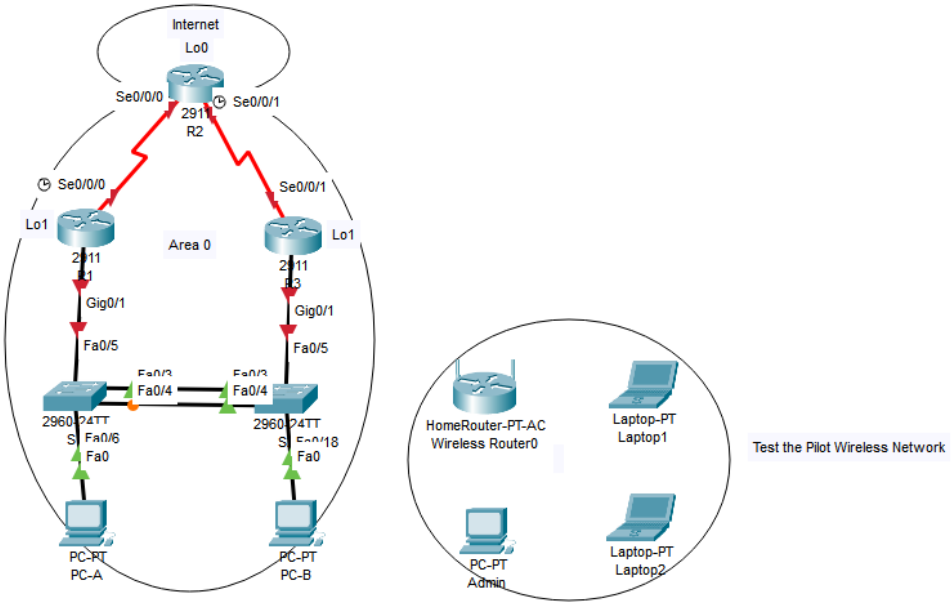
Technological University Dublin
Blanchardstown Campus

Topology Diagram



1. Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/1	172.27.0.1	255.255.255.0	N/A
	S0/0/0	172.27.123.1	255.255.255.252	N/A
	Lo1	172.27.1.1	255.255.255.0	N/A
R2	S0/0/0	172.27.123.2	255.255.255.252	N/A
	S0/0/1	172.27.123.5	255.255.255.252	N/A
	Lo0	209.165.200.225	255.255.255.248	N/A
R3	G0/1	172.27.0.3	255.255.255.0	N/A
	S0/0/1	172.27.123.6	255.255.255.252	N/A
	Lo1	172.27.4.1	255.255.255.0	N/A
S1	VLAN 1	172.27.0.11	255.255.255.0	172.27.0.2
S2	VLAN 1	172.27.0.12	255.255.255.0	172.27.0.2
PC-A	NIC	172.27.0.21	255.255.255.0	172.27.0.2
PC-B	NIC	172.27.0.22	255.255.255.0	172.27.0.2

2. Assessment Objectives

Part 1: Configure Device Basic Settings (52 points, 45 minutes)

Part 2: Configure LAN Redundancy and Link Aggregation (19 points, 10 minutes)

Part 3: Configure OSPFv2 Dynamic Routing Protocol (20 points, 30 minutes)

Part 4: Verify Network Connectivity and HSRP Configuration

Part 5: Configure the pilot Wireless Network (9 points, 10 minutes)

1. Configure Device Basic Settings

    1. Configure R1.

Configuration tasks for R1 include the following:

Configuration Item or Task	Specification	Check
Disable DNS lookup		
Router name	R1	
Interface G0/1	Set the Layer 3 IPv4 address. Refer to the Addressing Table for IPv4 address information. Activate Interface	
Interface S0/0/0	Set the Layer 3 IPv4 address. Refer to the Addressing Table for IPv4 address information. Activate Interface	
Interface Loopback 1 (LAN)	Set the Layer 3 IPv4 address. Refer to the Addressing Table for IPv4 address information.	

2. Configure R2.

Configuration tasks for R2 include the following:

Configuration Item or Task	Specification	Check
Disable DNS lookup		
Router name	R2	
Interface S0/0/0	Set the Layer 3 IPv4 address. Refer to the Addressing Table for IPv4 address information. Activate Interface	
Interface S0/0/1	Set the Layer 3 IPv4 address. Refer to the Addressing Table for IPv4 address information. Activate Interface	
Interface Loopback 0 (Simulated Internet connection)	Set the Layer 3 IPv4 address to 209.165.200.225/29.	
Default route	Configure a default route out Lo0.	

3. Configure R3.

Configuration tasks for R3 include the following:

Configuration Item or Task	Specification	Check
Disable DNS lookup		
Router name	R3	
Interface G0/1	Set the Layer 3 IPv4 address. Refer to the Addressing Table for IPv4 address information. Activate Interface	
Interface S0/0/1	Set the Layer 3 IPv4 address. Refer to the Addressing Table for IPv4 address information. Activate Interface	
Interface Loopback 1 (LAN)	Set the Layer 3 IPv4 address. Refer to the Addressing Table for IPv4 address information.	

4. Configure S1.

Configuration tasks for S1 include the following:

Configuration Item or Task	Specification	Check
Disable DNS lookup		
Switch name	S1	
Assign an IPv4 address to the default SVI.	Refer to the Addressing Table for IPv4 address information.	
Assign the default-gateway.	Refer to the Addressing Table.	
Force trunking on interfaces connected to S2.	Use VLAN 1 as the native VLAN.	

5. Configure S2.

Configuration tasks for S2 include the following:

Configuration Item or Task	Specification	Check
Disable DNS lookup		
Switch name	S2	
Assign an IPv4 address to the default SVI.	Refer to the Addressing Table for IPv4 address information.	
Assign the default-gateway.	Refer to the Addressing Table.	
Force trunking on interfaces connected to S1.	Use VLAN 1 as the native VLAN.	

6. Configure IPv4 addresses on PCs.

Configuration Item or Task	Specification	Check
Configure static IPv4 address information on PC-A	Refer to Addressing Table for IPv4 address information.	
Configure static IPv4 address information on PC-**B**	Refer to Addressing Table for IPv4 address information.	

Part 1 score = 52/100

2. Configure Redundancy and Link Aggregation

    1. Configure HSRP on R1.

Configuration tasks for R1 include the following:

Configuration Item or Task	Specification	Check
Configure the HSRP virtual IP address on interface G0/1. Make this the primary HSRP router. Configure so this router becomes the primary HSRP router on a reboot. (priority 150)	Group: 1 Virtual IP address: 172.27.0.2	

2. Configure HSRP on R3.

Configuration tasks for R3 include the following:

Configuration Item or Task	Specification	Check
Configure the HSRP virtual IP address on interface G0/1.	Group: 1 Virtual IP address: 172.27.0.2	

3. Configure an LACP EtherChannel between S1 and S2.

Configuration tasks include the following:

Configuration Item or Task	Specification	Check
On S1, configure an LACP EtherChannel on interfaces connected to S2.	Use group 1 and enable LACP unconditionally.	
On S2, configure an LACP EtherChannel on interfaces connected to S1.	Use group 1 and enable LACP only if a LACP device is detected.	
On S1 and S2, enable rapid spanning-tree.		
Enable portfast and bpdu guard on S1 and S2 hosts.	Use In fa0/6 on S1 and fa0/18 on S2	

Part 2 score = 71/100

3. Configure OSPFv2 Dynamic Routing Protocol
    1. Configure OSPFv2 on R1.

Configuration tasks for R1 include the following:

Configuration Item or Task	Specification	Check
OSPF Process ID	1	
Router ID	1.1.1.1	
Advertise directly connected networks.	Use classless network addresses. Assign S0/0/0 and G0/1 and lo1 interfaces to Area 0.	
Set all LAN interfaces as passive (G0/1, Lo1)		
Change the default cost reference bandwidth to support Gigabit interface calculations.	1000	

2. Configure OSPFv2 on R2.

Configuration tasks for R2 include the following:

Configuration Item or Task	Specification	Check
OSPF Process ID	1	
Router ID	2.2.2.2	
Advertise directly connected networks.	Use classless network addresses. All connected networks should be assigned to Area 0 except the Lo0 network.	
Propagate the default route to all other OSPF routers.		
Set all LAN interfaces as passive (Lo0)		
Change the default cost reference bandwidth to allow for Gigabit interfaces.	1000	

3. Configure OSPFv2 on R3.

Configuration tasks for R3 include the following:

Configuration Item or Task	Specification	Check
OSPF Process ID	1	
Router ID	3.3.3.3	
Advertise directly connected networks.	Use classless network addresses Assign S0/0/1 and G0/1 and Lo1 interfaces to Area 0	
Set all LAN interfaces as passive (G0/1, Lo1)		
Change the default cost reference bandwidth to support Gigabit interface calculations.	1000	

4. Verify Network Connectivity and HSRP Configuration
    1. Verify end-to-end connectivity.

Take corrective action if results are other than expected.

From	Command	To	Expected Results	Check
PC-B	ping	PC-A	Ping should be successful.	
PC-B	ping	Default Gateway	Ping should be successful.	
PC-B	ping	209.165.200.225	Ping should be successful.	
PC-B	tracert	209.165.200.225	Trace should route through R1.	

Part 4 score = 91/100

5. Configure the Wireless Clients

Configuration Item or Task	Specification	Check
Connect Admin to WR	Use straight through cable to G0/1 on WR	
Configure Admin to use DHCP	To reach the **WR** management page, **Admin** must communicate on the network. Enable DHCP	
Connect to the WR Web Interface	Login into the routers web interface using the gateway IP. Use **admin** for both the username and password.	
Configure the Internet Port of WR	**Static Assignment As follows;** **Internet IP Address**: 209.165.200.225 **Subnet Mask**: 255.255.255.252 **Default Gateway**: 209.165.200.226 **DNS Server**: 209.165.201.1 **Save Settings**	
Configure the Wireless Settings	Only configure the wireless settings for 2.4 GHz **(SSID)** to **aCompany** **Standard Channel** to **6 - 2.437GHz** **Save Settings**	

Configuration Item or Task	Specification	Check
Configure wireless security settings	Wireless security settings using WPA2 select **WPA2 Personal** for the Security Mode keep the default **AES** setting **Cisco123!** as the passphrase. Save Settings	
Connect the Wireless Clients	Laptop1 and Laptop2 Enable DHCP Test for Connectivity between L1 and L2	

Part 5 score = 91 – 100

1.